EXPERIMENT : 9

AIM:

Experiment to understand Matplotlib library use cases in Data Science through visualization

DESCRIPTION:

Use Iris data set to understand the Matplot lib library

CODE:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

%matplotlib inline

data = pd.read_csv('/content/sample_data/iris.csv')

data.info()
print(data.describe())

print(data.value_counts('species'))

sns.countplot(x='species', data=data)
plt.show()

dummies = pd.get_dummies(data['species'])

FinalDataset = pd.concat([
```

```
    dummies,
    data.iloc[:, [0, 1, 2, 3]]
], axis=1)

print(FinalDataset.head())

sns.scatterplot(
    x='sepal_length', y='sepal_width',
    hue='species', data=data
)
plt.show()

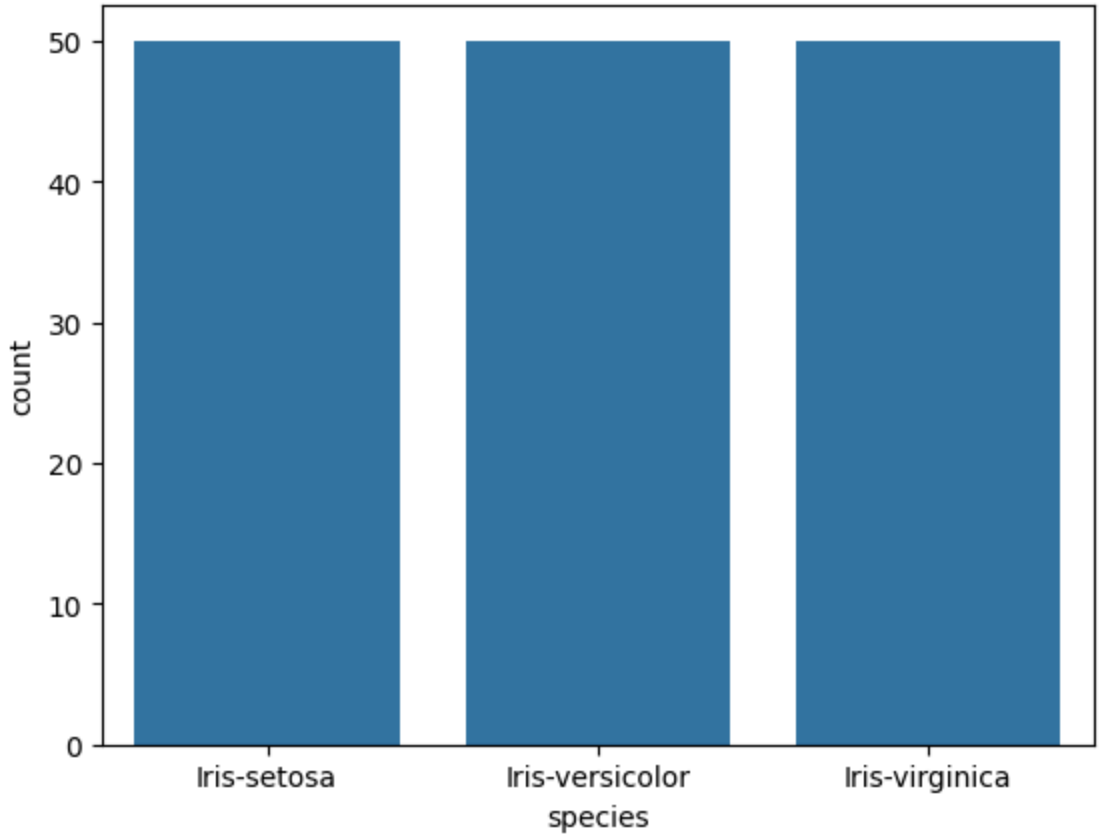
sns.scatterplot(
    x='petal_length', y='petal_width',
    hue='species', data=data
)
plt.show()

sns.pairplot(data, hue='species', height=3)
plt.show()

for column in ['petal_length', 'petal_width',
               'sepal_length', 'sepal_width']:
    sns.FacetGrid(data, hue='species', height=5).map(
        sns.histplot, column
    ).add_legend()
```

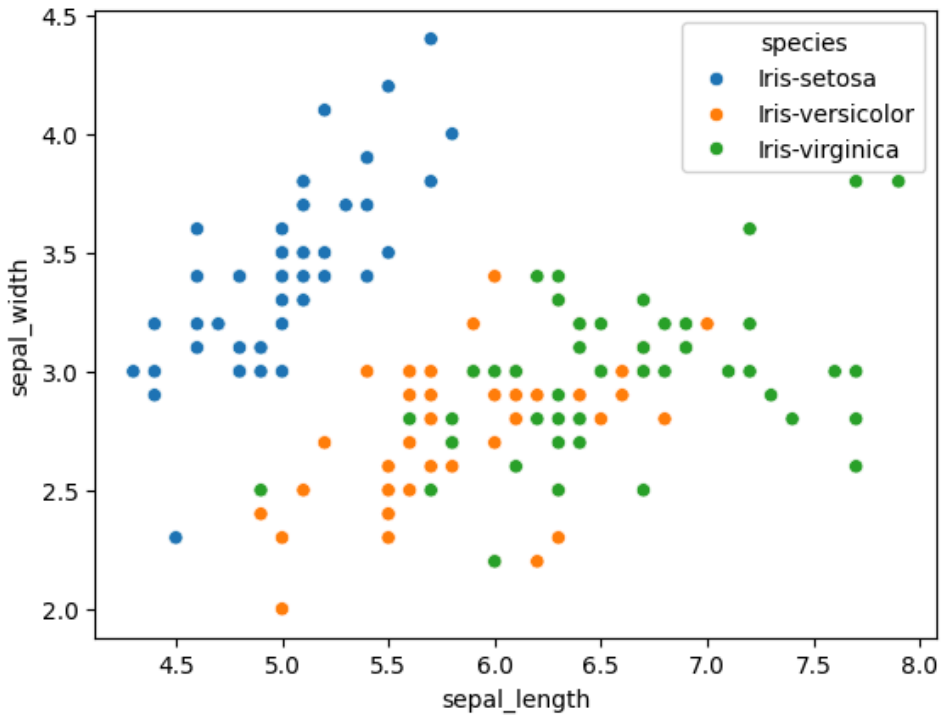
OUTPUT:

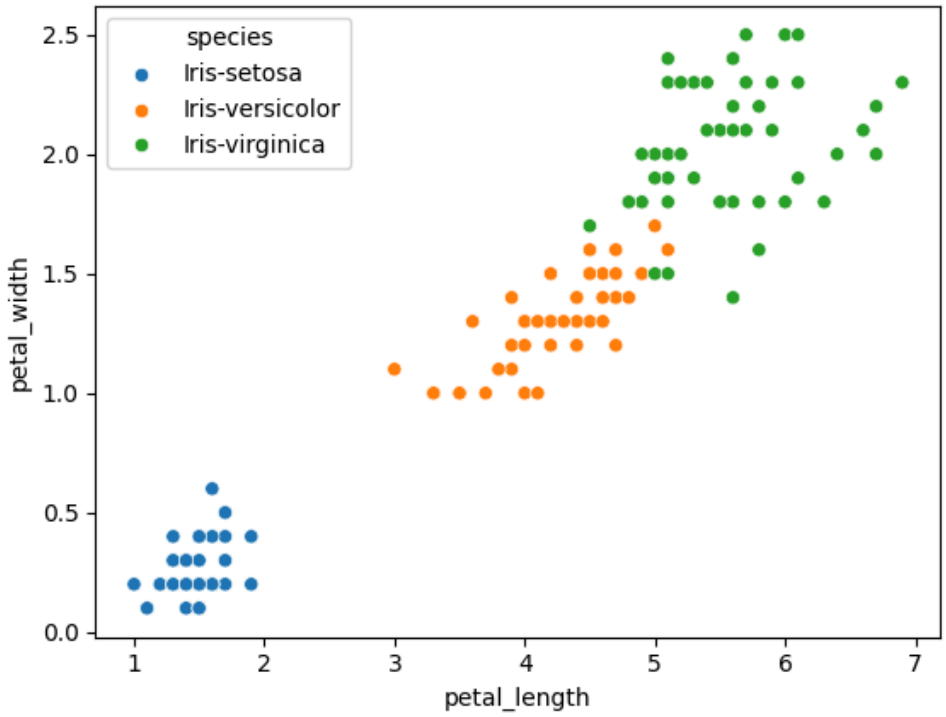
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    object 
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
       sepal_length  sepal_width  petal_length  petal_width
count    150.000000   150.000000    150.000000   150.000000
mean       5.843333     3.054000      3.758667     1.198667
std        0.828066     0.433594      1.764420     0.763161
min        4.300000     2.000000      1.000000     0.100000
25%        5.100000     2.800000      1.600000     0.300000
50%        5.800000     3.000000      4.350000     1.300000
75%        6.400000     3.300000      5.100000     1.800000
max        7.900000     4.400000      6.900000     2.500000
species
Iris-setosa        50
Iris-versicolor    50
Iris-virginica     50
Name: count, dtype: int64
```

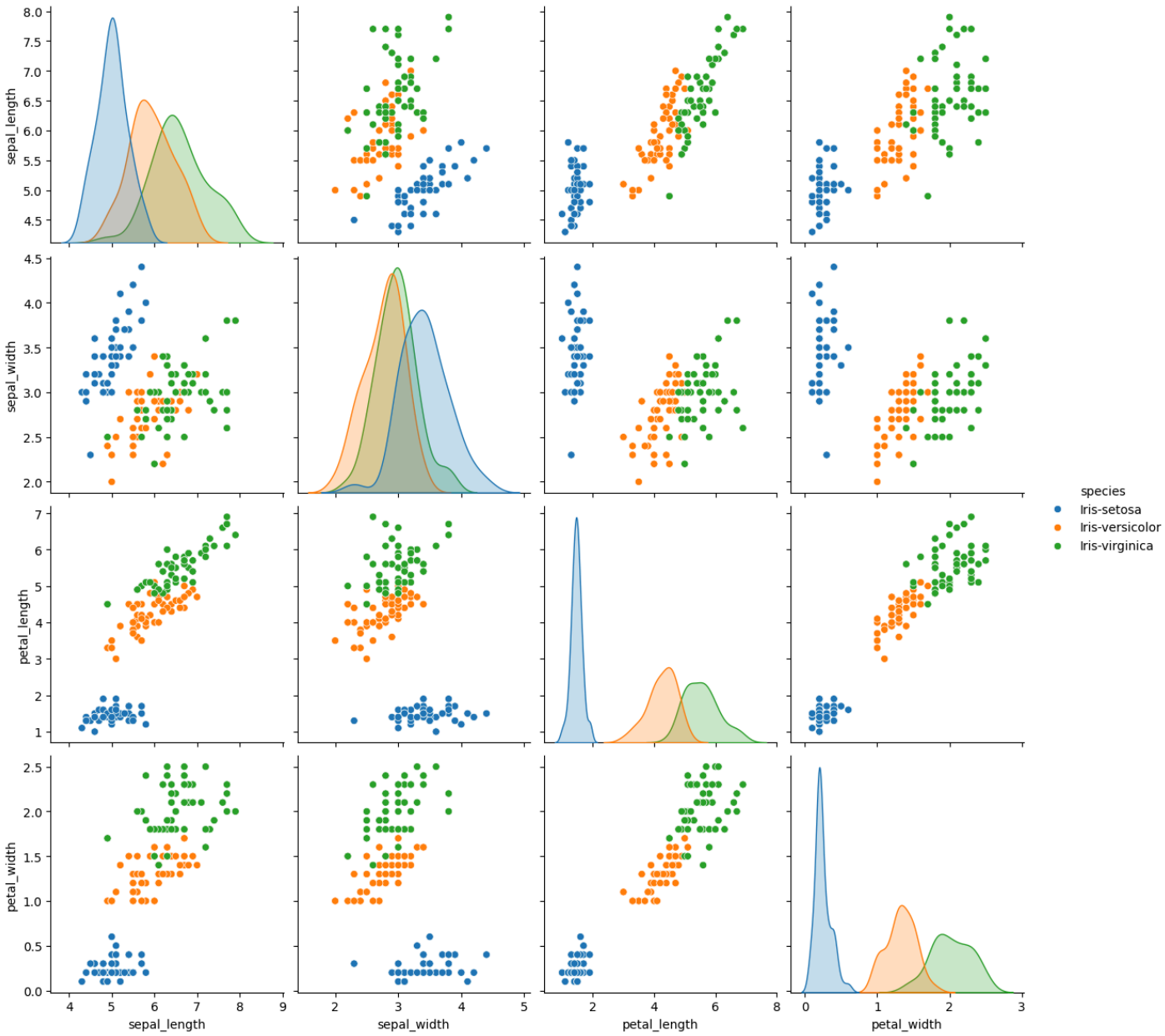


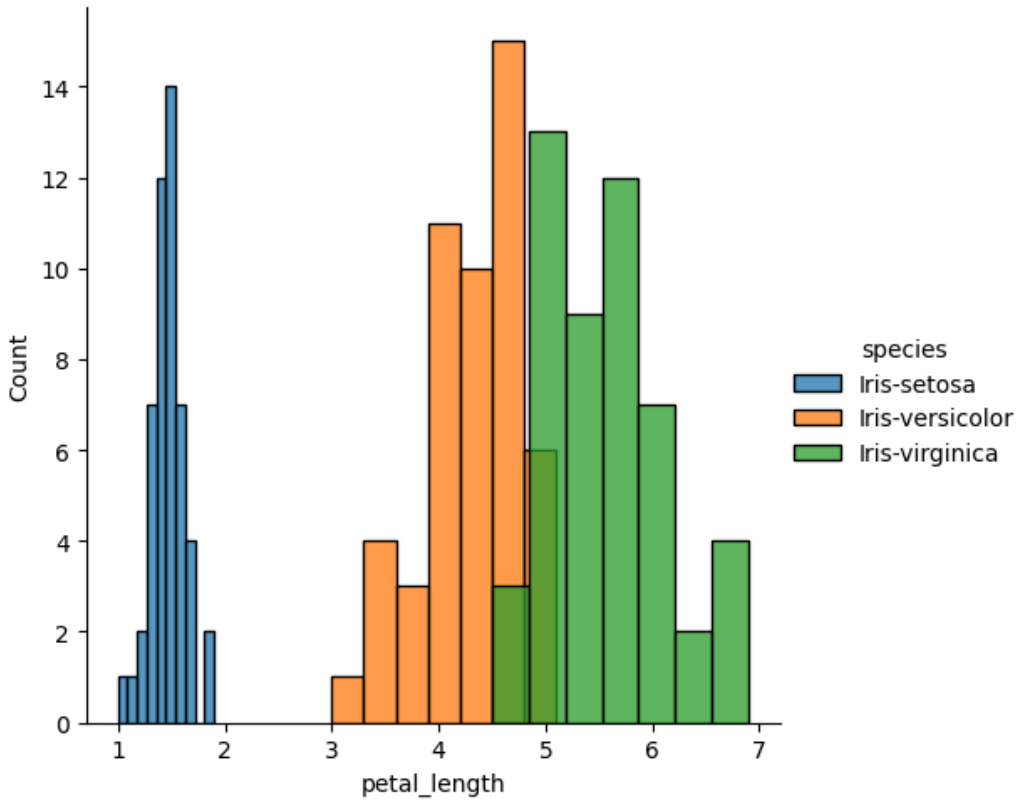
	Iris-setosa	Iris-versicolor	Iris-virginica	sepal_length	sepal_width	\
0	True	False	False	5.1	3.5	
1	True	False	False	4.9	3.0	
2	True	False	False	4.7	3.2	
3	True	False	False	4.6	3.1	
4	True	False	False	5.0	3.6	

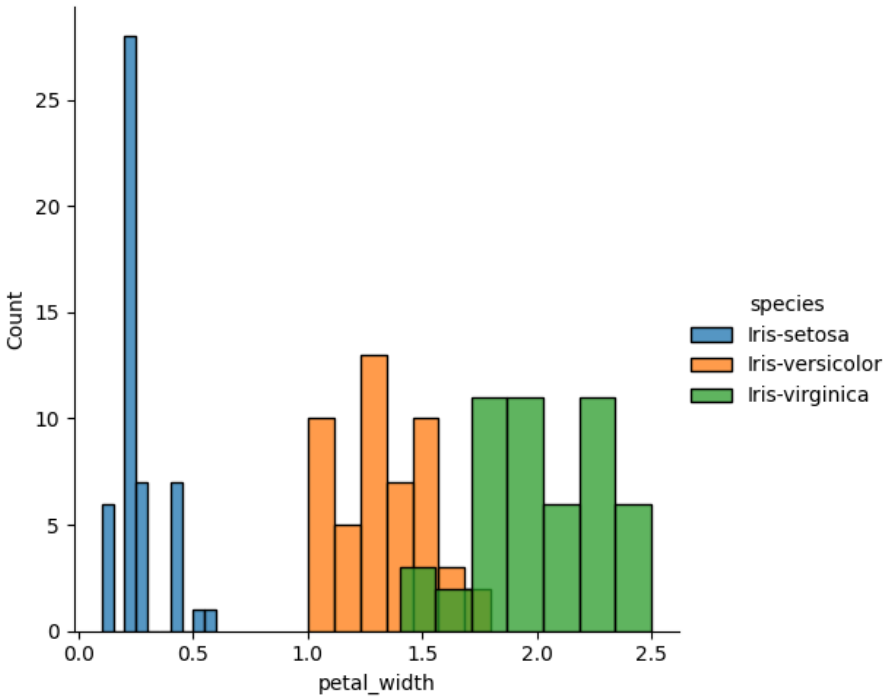
	petal_length	petal_width
0	1.4	0.2
1	1.4	0.2
2	1.3	0.2
3	1.5	0.2
4	1.4	0.2

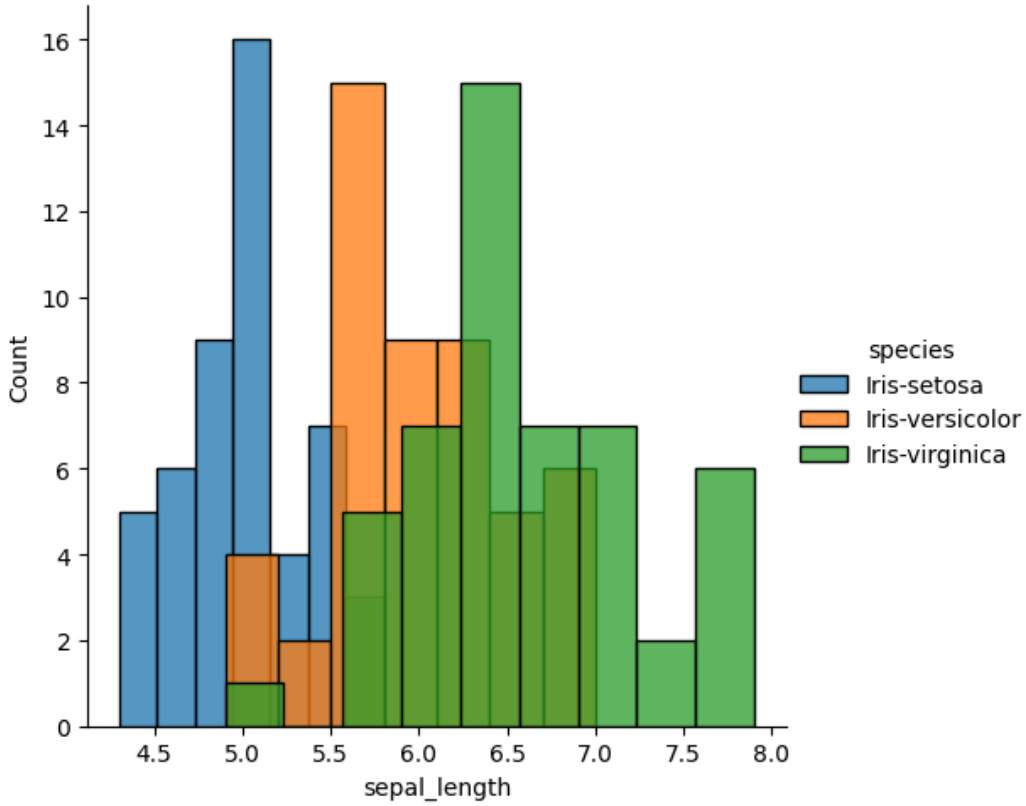

species
Iris-setosa
Iris-versicolor
Iris-virginica
sepal_width
sepal_length
4.5
4.0
3.5
3.0
2.5
2.0
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0

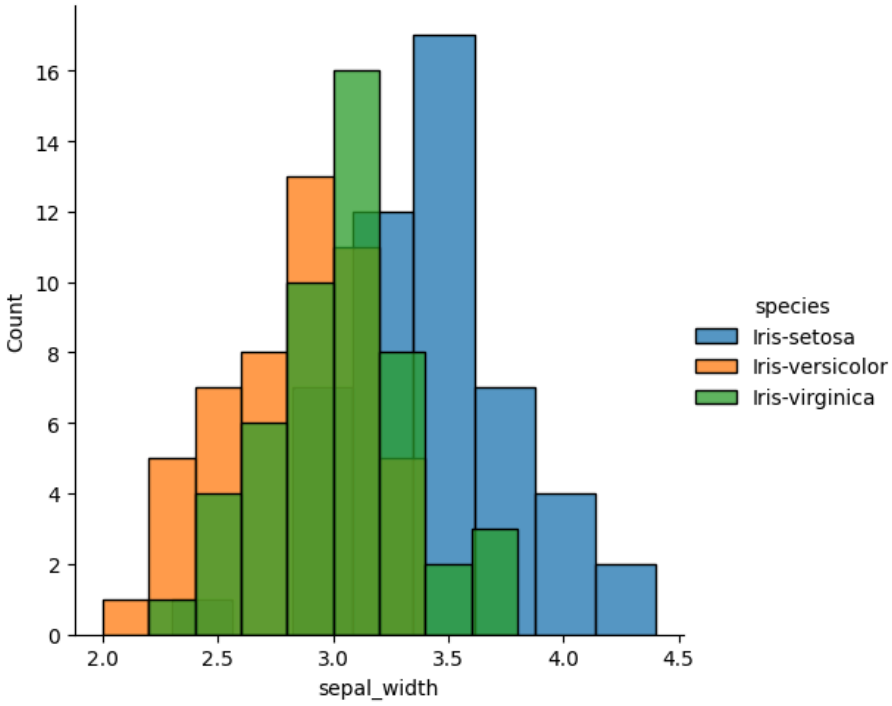

species
Iris-setosa
Iris-versicolor
Iris-virginica
petal_width
petal_length
2.5
2.0
1.5
1.0
0.5
0.0
1
2
3
4
5
6
7


sepal_length
sepal_width
petal_length
petal_width
species
Iris-setosa
Iris-versicolor
Iris-virginica


Count
petal_length
species
Iris-setosa
Iris-versicolor
Iris-virginica


Count
petal_width
species
Iris-setosa
Iris-versicolor
Iris-virginica


Count
sepal_length
species
Iris-setosa
Iris-versicolor
Iris-virginica


Count
sepal_width
species
Iris-setosa
Iris-versicolor
Iris-virginica

RESULT:

The Matplotlib library was successfully used to visualize the Iris dataset through different types of graphs. The visualizations helped understand data distribution, identify patterns, and compare the features of different Iris species.